

ActInf Livestream #008.1: "Scaling active inference" (2019)

Livestream | 1:21:41

hello and welcome everyone to the active inference live stream this is act in stream 8.1 it is November 10th 2020 welcome to tecom everyone we are an experiment in online team communication learning and practice related to active inference you can find us at our website active inference org you can find us at our Twitter at email our public keybase team or on YouTube this is a recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and as far as video etiquette for live streams go um mute yourself if there's noise in your background raise your hands so that we can hear from everybody on the stack and use respectful speech Behavior ET today we are excitingly in act mstream 8.1 and it's going to go a little something like this we're going to start with some intros and warm-ups and then we'll get to the sections of 8.1 in 8.1 as well as 8.2 we're going to be discussing the paper scaling active inference by chance at all in 2019 thanks to Blue for the suggestion of the paper we're going to go through the goals the abstract and the road mapap and then we'll have a little notation and math overview we'll also take a look at the figures with a special focus on what each figure is showing or representing or

invoking in today's discussion 8.1 and next week's discussion 8.2 we're going to be discussing this same paper so save and submit your questions and also get in touch with us if you want to participate for the rest of 2020 we're going to be discussing this paper 8 as well as 9 10 and 11 would you can learn more about by checking our Twitter so check it out there all right here we are in the intros welcome again Shannon I'll kick out other

Shannon so for the introductions we'll go around and just introduce yourself and your location and we'll uh roll with it as people gain and uh lose the chat so just say hello in a short introduction and then pass it to somebody else I'm Daniel I'm in California and I will pass it to Blue hi I'm blue and I am in Southern New Mexico in Las cruus and I will pass it to

Sasha hi I'm Sasha I'm also in California and um I will pass it to Ivon hello my name is Ian I'm in Moscow Russia and I pass it to Alejandra I think she disappeared for a second so let's go to Alex

okay hello I'm Alex I'm also in Moscow Russia so welcome

everybody cool and Alejandra has been on before so hopefully we know her but when she jumps back in we can hear from her so for these warm-up questions and also it's pretty cool to have a slightly smaller panel so I'm really looking forward to hearing everyone's

perspective on what they got out of the paper what they brought to the paper it'll be really cool to hear about so the first question is what made you excited to read this paper and or to learn about these topics so while people are raising their hand and feel free to do so something that initially made me excited to read this paper was just about learning how active inference could be applied to the kinds of problems that people apply other machine learning algorithms to so that's a starting point blue so I picked this paper I um hadn't like thoroughly like digested it when I when I picked it I had read um the first author's other paper in PL computational biology but like applying um like agent based models and like modeling parameters to uh active inference was was like really what intrigued me and the um this paper I especially liked because of the ability to apply it in a continuous frame framework as as opposed to like something in discret time steps and I thought that that's that was significant and important cool welcome Alejandra do you want to just give your quick intro and say hi oh never mind uh anyone else want to add something that made them excited to learn about this topic or read about the paper yeah thanks again for the suggestion blue the second question is what is your experience or background with computers SL comput ational

modeling and what is something interesting that you've been learning about or studying recently in the area and I think that my goal with this question was to show to our listeners SL viewers that there's many computational backgrounds that intersect in active inference so it's a community that includes some people who are really on The Cutting Edge of the computational modeling and that's what we see reflected in this paper so big props to everybody who is reading and talking about it because it is at The Cutting Edge of that type of machine learning research but also there's this deep tradition of action-oriented research and embodied cognition and so many other areas that might not necessarily entail a computational background or even a computational approach in how they're studied until this type of intersectional research so does anyone want to just something that they've been learning about in computers or what type of computer modeling have they had experience with so I can start a lot of my computer work had been in bioinformatics which does deal with computers but it's very different kinds of questions than the machine Learning Community however increasingly there's application of machine learning to bioinformatics so this was just kind of like a different perspective on machine learning outside of the biological contexts that I had mainly seen it in and I'll just throw up the third

question in case anyone wants to answer
that one which is what kinds of problems
might machine learning based upon active
inference be applied to so you can just
raise your hand if you yep go ahead

Sasha um yeah I would say I guess to
some of all these questions is I have
very limited experience and working with
computers uh but my background is in
neuroscience and I'm always curious to
to um learn what uh machine learning um
has to offer for understanding the
process of learning and how that can
link back to neurobiology Concepts um
and so I'm interested in in
understanding these equations and having
uh a better grasp on um what the
relationships are

like cool because learning is really a
biologically inspired topic it's based
upon learning in organisms or the kind
of control that organisms enact in their
environment and that's not even getting
into the whole neural network SL actual
brain debacle so definitely these kinds
of learning questions any other thoughts
on this

one

blue so like you my background is like
bioinformatics but I've like branched
out since like getting out of doctoral
Research into like doing all kinds of
different

computational and agent-based modeling
type applications even Quantum Computing
a little bit um this system particularly
um interests me because like Sasha I'm
interested in learning like how things

learn but also like from the Beyond
brain perspective like stay in biology
but think about biological systems just
at a very simple level like slime molds
or something like this how do these
systems learn right because still
there's some element of learning that
happens and when you can kind of
deconstruct it into like a
computational like model then you can
start to like recognize input output
into simpler biological systems beyond
the
brain cool and let's return to that
multiscale cognition when we're a little
bit later on but I wrote that down so we
can definitely return there any last
thoughts on the warm-ups before we get
get into the
paper cool all right so today we are
going to be talking about this scaling
active inference paper which is an
archive paper and the paper presents its
goal as or what it does as we present a
model of active inference that is
applicable in high-dimensional control
tasks with both continuous States and
actions our model builds upon previous
attempts to scale active inference by
including an efficient planning
algorithm as well as the quantification
and active resolution of model
uncertainty our model makes two primary
contributions first we showed that the
full active inference construct can be
scaled to the kinds of tasks considered
in the reinforcement learning literature
this involved extending previous models

of deep active inference to include
model uncertainty and expected
Information Gain second we highlighted
the overlap between active inference and
state-of-the-art approaches to
model-based reinforcement learning
welcome back Alejandra and some of these
topics were introduced or at least
initially brought up in the ACT imp
stream 8.0 so if you're a little bit
just unsure what is modelbased versus
non-m modelbased reinforcement learning
or what is reinforcement learning or
what is machine learning it's all good
there's kind of a sequence of ideas that
you can follow to understand why this
paper is so exciting and just go to ACM
stream 8.0 if you're curious the way
that we can phrase what the Big Goal or
question of the paper is in terms of non
active inference uh vocabulary would be
how can we apply active inference to
challenging control tasks and like blue
mentioned these are the ones that often
involve continuous decisions being made
so like turning up a dial um or down or
steering wheel rather than like chess or
Checkers or Connect 4 or the Mario Kart
example which now I will never forget
and then how to connect these continuous
challenging control tasks which have
elements of planning and all these other
things happening and connect these tasks
formally to reinforcement
learning any thoughts or questions on
the goal of the
paper cool so the abstract which we'll
just run through is in reinforcement

learning RL agents operate in partially observed and uncertain environments isn't that the truth modelbased reinforcement learning suggests that this control of these partially observed and uncertain environments is best achieved by learning and exploiting a probabilistic model of the world so just to be clear this is what traditional model-based reinforcement learning is model free or unsupervised reinforcement learning would be learning the very very direct relationships between outcomes like reward and States modelbased reinforcement learning abstracts a little bit above that and it considers how learning occurs on a probabilistic model of the world active inference is an emerging normative framework in cognitive and computational Neuroscience that offers a unifying account of how how biological agents achieve this control in this framework inference learning and action emerge from a single imperative that maximizes the basian evidence for a niched model of the world and so learning inference and action are all part of the organism's generative model and we're going to see how that gets specified and how it plays out the challenge is that implementations of this process this integration between inference and action if you will have thus far been restricted to low-dimensional and idealized situations so this could be for example playing a very simple game or doing a 2 by two Matrix like a

prisoners dilemma or there are other cases where active inference has been used for slightly higher dimensional but still quite idealized situations like reading from a known set of options here we present a working implementation of active inference that applies to higher dimensional tasks with proof of principal results demonstrating efficient exploration and an order of magnitude increase in Sample efficiency over strong model-free baselines so just by reading that we can look forward to some improvements on classical algorithms and as far as this magnitude increase in Sample efficiency the reason why they're focused on Sample efficiency is because these distributions that we want to understand the shape of are not just like a simple curve there's a lot of Dimensions to them and they're very rugged so small changes can change your height so to speak quite a lot which means that the challenge is to sample effectively not get trapped in one area or not to over sample one region at the cost of another our results demonstrate the feasibility of applying active inference at scale and highlight the operational homologies between active inference and current model-based approaches to reinforcement learning so what would be the implications of that well it'd be pretty epic if all of the hype and Buzz that we hear about machine learning or reinforcement learning or all the places where we know reinforcement

learning gets applied ranging from recommendation algorithms to maps to just all these types of things could active inference play a role in that and what what would that do for our conception of these kinds of algorithms and there's probably a lot of directions to take that but that's the equivalence is if there is a homology and potentially even a superiority of active inference techniques in certain cases over reinforcement learning then it liberates us from some of the baggage of the straightforward model-based uh reinforcement learning or the critiques of reinforcement learning but also opens the door to some new ways to implement strategies and understand the systems that we're working with cool so the paper has a pretty clear road map this is like driving on Highway 10 in New Mexico and there's a introduction section uh the relationship to further work is actually section five which we'll get to in a second but the paper begins with an introduction and a consideration of active inference as a topic it then moves directly to the model and lays it out in the six subsections 3.1 through 3.6 from a mathematical perspective we're going to be focusing primarily on one two and three because those are the parts that we've heard the most about and they're the most relatable so the generative model and the recognition distribution are the p and the Q mind your p 's and

q's and this is everything that we've been talking about with the two directions between the agent and the world and whether we think about that in the basian structural computationalist perspective or the an activist perspective the whole tale of two densities integrating internalism and externalism all these things come together in 3.1 3.2 learning and inference is where we're going to draw the bow together with reinforcement learning and about other machine learning algorithms

3.3 is where we're going to come closer to control theory because we're going to be talking about setting policies for Action then 3.4 56 hopefully when Alec is here next week we'll have a little bit uh more detail there maybe we could even see a simulation or hear about some of the degrees of freedom that the authors had in the model why they chose to do what they did the part that will be very cool as an experimental biologist is the experimental section and here it's an interesting approach to do one set of tasks that conveys exploration of a state space and another set of tasks that conveys exploitation so they're saying like this model has two hands you know it's good at exploring but it's good at exploiting as well then there's the connection to previous work and the three areas that they tie it most directly to with the keywords as well as in this closing section are related to deep active

inference which in contrast to I guess shallow active inference would mean that you're not just minimizing your surprise about your instantaneous sensory information but you have a deep or a counterfactual model that goes through time so that's like the example of taking a jacket even though you're not cold yet that's deep active inference you're reducing your surprise about future states of your temperature under a generative model where if you go outside and it's cold you're going to get cold but shallow active inference would be like you wouldn't put on the jacket or you wouldn't even take it with you because it wouldn't be a relevant affordance in the one second time frame but it actually is relevant in the 1second time frame under a deep generative model of temperature the second two are model-based reinforcement learning which we've mentioned a few times and Information Gain information is reduction of uncertainty and when we're reducing our uncertainty about environmental causes it helps us plan better actions but we also need to reduce our uncertainty about which actions to take and that's called policy and that's what Rel relates us here to control theory then there's a discussion and conclusion okay before we jump into the math in the actual sections are there any thoughts or questions on the road

map all right here is a partial notation reference and I just separate them onto

two sides and so if you're only going to look at a few parameters uh if your parameter overload hits in at six the ones on the left are the important ones these are the ones that will help you get 80% of what is happening and then the next 10% are the ones on the right and then there's a few other ones which aren't on this page that are more like the esoteric statistical parameters the big parameter is s with a hat just like Sasha with a hat and that's the true state of the environment so s with a hat is what is actually trying to be predicted and that's like the actual temperature of the environment or the actual amount of sunlight but that's not what the agent is able to track directly it's able to make an estimate of the environmental State and you can imagine there's times where that's an easy estimate to make there's times where it's a hard estimate to make and then this is the difference between the discrete and the continuous case would be if the state that you're estimating is is it day or night it's a zero or one that's a discrete case a continuous case would be like what is the amount of light and that's a continuous variable so there's a lot more options a lot more Shades of Gray infinite of them that's s is the agent's internal ongoing estimate of this environmental state state that it's trying to track and we can think about it with just one s or there can be multiple s 's like is it humid or is it dry that's one axis another one is it

light or dark o are the observations so in the case of light being estimated that would be like the observations of light on the retina in the case of temperature it's like a thermometer reading a are the policy actions by the agent and those are arising from π which is a policy and a policy is like a layer above action so for example action would be moving my left foot my policy would be if I hear this sound then I move my left foot another policy would be I always move my left foot and so there's different kinds of policies that can be very nuanced they can be conditional they can be related to each other or they can be very straightforward and Standalone t with italics is the time steps of the model so this is just a modeling parameter that's going to come up again and again and it has to do with how time is measured in terms of epochs or states or continually okay then just to the right side we have little and big Θ and little Θ is often used just to mean a parameter of the generative model and the bigger Θ is like the space of all the parameters so those are kind of used similarly then Λ is related to the likelihood distribution it relates some of the statistical parameters and then f is the free energy and this represents like if you could be on this super smooth variational optimization landscape just with just as if you would really want to be following $\hat{s}$ but you actually have to follow s with free

energy if we could be following F then we'd be doing perfectly you just strictly cannot do any better than that however we can't directly track F so we have to track G which is the expectation of free energy which we'll also be able to look at okay any thoughts or questions on notation anyone seeing similar notation to another field or a difference is there something that we want to that you'd imagine that a control theory model would have that isn't in

here all right anyone can just raise their hand at any time I'm not sure where several of our colleagues have gone

to let's start by thinking about the generative model so this is a generative model P of a bunch of variables and this is going to be kind of like what the organism is slash does we're going to leave for a second some of these uh avenues that can be arising with the niche symmetry which we've talked about in the context of yellow bromberg's and actual constants work and there's the a lot of other things that we've brought up let's just try to isolate what's really relevant which is that the agent is doing some type of implementation or is some kind of an implementation of a generative model of a couple of things at once it's generating a model of observations through time the Tilda on top of a variable means through time so observations through time and States through time you can think of as a

function of policy choices π and parameters of the model Θ so for example one of my parameters in my Θ is like um if I get hit by a bike then it's going to hurt and then my generative model is like under the policy of looking both ways to cross the road I predict that my States will include not being hit by a bike and so I won't have any pain observations under that same policy you could imagine um other types of state and observation predictions and then there are a few subsequent mathematical details about the insides of this equation it's a little bit bracketed no pun intended in the top and that's a shorthand and then we can EXPLORE and unpack it a few more levels so like on that first row we can see that that that $p(\pi, \Theta)$ those um uh relationships can be unpacked on the other side of the equal sign the right hand side to like just the probability of the parameters existing and then just the probability of the policy existing multiplied by this big other thing so why would that be the case and we won't go through all of them in this detail but I think this one's really helpedful to understand for active inference and people just raise your hand if you want to talk but just to keep it like informational and fun I'll just keep going until somebody wants to ask a question so the reason why $P(\pi)$ is there the probability of the policy is like if the policy is unrealistic you don't even want to

consider it you want to basically take unlikely policies impossible or implausible policies and systematically not consider them so if I'm trying to get out of the way of the incoming bike and I'm spending a lot of my time or my clock Cycles or a lot of my cognitive infrastructure on action strategies that involve me doing a Spider-Man and swinging from a building it's it's so implausible that it's probably not going to be effective so we want to have something in our model that represents likely or possible policies being considered a little bit more richly than implausible policies and similarly for p of data which is a little more abstract because it's just a parameter of the model but it's like saying if it's an unlikely parameter State it's also not worth thinking too much about I don't need to plan for it being -40 in California however our Russian colleagues might need a plan for that so it's going to be Niche dependent and contextual whether they need to plan for it or not okay now let's go into this big pie uh notation on the right side of the right hand side of the first line of equations this is kind of like a sigma Sigma adds up the elements of a series so like Sigma of you know $1 + \frac{1}{2} + \frac{1}{4} + \dots$ or something and the π is like a big multiplier and it's saying from the first time step T equals 1 through all of the time steps Big T do this calculation what is the calculation well it's the probability of

observations given States so and that's observation out a certain time given the state at that time so it's like if this my estimate of the state not s hat just s with hat if my observ if my state is getting hit by a bike there's going to be a very likely observation of it hurting now that doesn't say whether it's likely that I'm getting hit by a bike it's just that it is the case if it is that state then there's going to be this observation so that's the mapping between observations and States but it's observations probability conditional with a straight line on the state being a certain way okay that's the first part and the second part is the probability of that state arising conditional on the previous state and the previous policy and the parameter and so again with this like bike model it's like what is the probability of my estimated State s oft at that time being hit by a bike given that I wasn't being hit by a bike before and my policy was looking both ways and my Theta model parameters tell me that if I look both ways it's really likely that I'll stay safe for example but stay safe isn't what's directly specified by the model Theta it's actually specifying the likelihood Ood that a given policy and state mapping are going to relate into the future with a certain State and then separately we can consider What observations arise given a state so it's a little bit like just showing our work or being very

clear with what is the actual state of
the world s hat what's the estimated
State s what are the um mappings between
policies States and future States that's
the right side then what's the mapping
between observ observations and states
in that first part okay any thoughts or
questions because the rest just kind of
continues from here in a similar way
um okay we'll just keep on you know
write down any cool questions that are
coming to mind Ivon yeah go
ahead well you're muted first but yes
then continue uh thank you uh could you
please explain one more time what's the
small Theta is it's parameter
of uh it's parameter
of the generative model but what is it
yes great question so we can actually
list we can go just just to show it in
the context of the
paper um and this by the way is in 3.1
but it introduces some of the other um
equations
earlier um
the Theta are the details of the model
that relate the transition distribution
so here we're looking at the states and
the previous policy the S_{t-1} and P_i of
 s_{t-1} and how given Theta it maps on
to S_t of so what is Theta just from
looking at the equation what is Theta
going to be telling us it's going to be
telling us how to map T_{t-1} 's
State and policy State estimate and
policy to the time T States so let's
just imagine that it were a linear model
so I had a linear $y = MX + b$ and that

was how we were going to map forward
just totally a simple way to do it then
Theta would be like a vector containing
m and
b or Theta would be the how many
dimensions of the model there are is how
tall Theta is going to be in a sense but
Theta are the parameters of the model
that say what the transition
distribution
is so uh yeah if it was a linear model
then Theta is kind of like a wrapper
around the parameters of the linear
model which is the slope M and then the
elevation B If This Were a neural
network with a million parameters then
Theta would have a million parameters
but it's kind of specifying the
likelihood that the S of tus1 and Pi of
T minus1 map on to S of T so let's just
say that Thea were one it just always
mapped to one for some reason then it's
like whatever these two are it's going
to map to one that's like you know $f(x)$
equals one yeah that's the parameter
that's the simplest possible model is
whatever the previous States and policy
are we just map to one the next one
would be $F(x)$ equals you know
insert your model there and so it is a
statistical point about what's a
parameter of a model and like aren't
these all parameters aren't they all
variables aren't they all random
variables aren't they all going to be
variables in my python code yes to all
of them and that's why there's a little
bit of nuance and that's actually why we

go into the details here because you can get pretty Tangled Up and Confused in the gray Zone with what's a estimator and what what is the what is the scientist estimating what is the organism estimating and a lot of times when we just look at it it's a little bit clearer and we can think about a few layers of unpacking with the top most level of understanding is this uh big p top equation then we can unpack it and Define well what does that really mean how do we pull out the likelihood of the parameters like the likelihood of theta from the likelihood of policy from this time dependent series how do we Define the distribution of P of O of T given s of T like the second line and so uh each of these we get continue unpacking and then each of them get specified in functions in programming which hopefully we'll hear from Alec about next week and there sometimes it becomes another level of uh Clarity when you look at the function definition and you can say oh this function is actually taking in variable a b and c and then instead of seeing them as just o $subt$ s $subt$ the variable's name could be States through time or there might be a more descriptive name so I know that especially if you haven't looked at these equations for a long time or uh perhaps many equations for a long time sometimes like you get halfway down the page it's like what what what is μ of theta again but it is referring to something specific it's just a little

bit of a shorthand to make sure that uh
it fits visibly in a containable space
so any other thoughts but good question
there with the
parameters so just what are the other um
yeah yeah Sasha go
ahead um yeah thank you that was really
useful to walk through those steps
especially with the bike example and it
um kind of brings to mind uh cases of
when something is very unlikely to
happen but the um but it's certain to
have uh really bad outcomes so like uh
you know it's really going to hurt if
you get hit by a bike versus um things
that are very likely to happen but don't
have
uh but are not certain of having a bad
outcome like I don't know um
stubbing your toe on something um and so
these are things that are weighted
differently in in a model uh based on
the certainty of uh and the severity of
the outcome so I think that's really
cool that this is all incorporated into
one model yeah and when we specify why
we're making certain decisions first
we're just laying these equations out to
describe our little control theory robot
playing Mario Kart or playing control
theory games on N64 but what if people
had a a little bit more nuanced
discussion on Collective decision-making
topics like somebody thinks well I think
it's totally unlikely that the minimum
miles per gallon of cars could be 40
miles by 2030 and someone else says
that's super likely actually it's

totally possible it's totally likely I think that it can be done or I think that this other policy that you're suggesting is implausible or that it will have a bad outcome and so people can disagree on whether a policy is actualizable or not for various reasons it might come across as an ethical constraint or legal constraint or logistical whatever it happens to be but for whatever reason that P of P_i is low for them and then we could unpack that and say okay all these different elements are there depending on the situation and then somebody else might say I'm with you on that outcome being unlikely I just don't think it's that big of a deal saying oh yep I think that that kind of a tsunami will happen once every hundred years but I think that my transition Matrix from tsunami damage to being repaired it just says that that's an easy transition so we don't really need to plan so heavily around it because when it does happen we'll be able to rebuild quickly for example so these just allow us to give a little bit more detail into our infrastructure really our architecture of decision-making and so I think blue that's definitely where we can take it back to this multiple scales of decision-making question just to look through a couple of these other equations before we move to Q here is the probability model of observations given States and so again that's like if the state my estimated State not s hat if my estimated state is

getting hit by a bike or stubbing my toe
the observations will be such and such
in my appropriate receptors and that is
given by a normal distribution which
means big n on the variable o of T and
semicolon parameterized by μ and σ^2
which are the mean and the variance
and then those together μ and σ^2
it turns out that those are the
hard parts to estimate just like the
problem or the challenge of doing a
linear regression on a billion data
points point would be estimating the
mean and the variance so you could fit
the mean by adding up all the numbers
and then dividing it by the amount of
numbers and for fitting the variance you
could use like a least squares model but
still that would be the computation that
you could do on pen and paper but would
just be very hard in this case we're
going to be doing what is called a
amortized learning or a functional a
functional is like a function of another
function and what we're going to be
doing is estimating these statistical
parameters
with this function Λ and then we're
going to fit the neural network about
that function Λ and so even though
neural networks are often understood as
providing additional degrees of
flexibility like being able to learn
conditional relationships between data
um this took me a little bit to think
about but the neural network is actually
just facilitating U s in this model it's
not saying that it would have to be this

way for all future ACM learning but it's actually still just fitting a neural network model of a normal distribution so we're finding the parameters μ and σ use it which are very difficult to estimate as you can imagine for nonlinear control task but we're still fitting it with a function around μ a normal so it's kind of this interesting juxtaposition of using Advanced Techniques like a neural network um High parametric techniques to actually drill back down to a mean and a variance and then so this is this p is this first part of this u_t series and then s_t of given the previous state and the previous policy is the second part here so this part here gets estimated with f of λ and then the uh second half of that series through time has another normal distribution with uh μ of θ and σ^2 of data and then those two get looped up into uh a functional over those two similarly like s_t of T is the right hand side what's being conditioned on and we're going to be estimating the mean and variance of what is conditioned on and then here it's similar we're making a function that is estimating what is being conditioned on okay then we have this uh early parts of the equation which is the probability of the policy is basically um it's going to be pretty simple in this case but this is where you could have a lot more degrees of flexibility this is sort of like an initial skeleton paper as far as what could be done with this framework and

that's kind of cool that's actually what defines it in some ways as being research and as being cutting edges this isn't the fine tuning of the parameters this is laying out the framework that's going to help us in a lot of other contexts and then the probability of the policy this is the part where we actually get to free energy okay so if we remember from the notation slide f is like the free energy where we strictly couldn't be making any better decisions if we knew that but G is the expected free energy so just like s is the $\hat{s}$ with the hat is real state of the environment that's like what we'd really want to track and free energy would be what our policy decisions would strictly want to be converging upon but we don't get $\hat{S}$ we get S and we don't get F we get G what is g where does it come into play well the probability of a policy is now here Σ is actually not this is Σ^2 a statistical parameter and it's kind of annoying but it is this way that this Σ is a soft Max function which basically is like the make a choice function in math it just is s for soft Max here Σ for variance here and what this is going to be is the negative expected free energy and sometimes there's some negatives and some logs getting thrown around and these are all just ways to really just say pick the best given the expected free energy of policy and so this is how we get to certain statements which have definitely been discussed on this uh

live stream before where it's like the agent picks the most likely policy and it's like wait but is it likely for me to win or lose what if it's a I know that it's unlikely for me to win or lose um if it's one in 100 that I'm going to win how is that me picking the most likely policy but that's actually what um is happening but it's not just simply the likeliest policy um there's a little bit more Nuance which we'll look into with G okay so that was sort of the P side let's maybe a little bit more quickly because we will recognize a lot of the variables and also a lot of the structural relationships between equations here's the CU so we have p and Q between the agent and the environment uh in a sense these are the two distributions that are linking it not that one is being computed by one or one is computed by the other but thinking about in that basian computationalist framework where we have like the data going in one model goes to the hyperparameters and then the hyperparameters generate leading to um the uh Hey Shannon welcome back if we if yeah um going back to the observed states of the world so Q um is going to be distinguished from p p is this implementation of the generative model of the world where Q is going to very distinctly have different parameters instead of P P_i Θ here we have q of s Through Time policies in Θ so this is a slightly simpler uh mathematical formulation and it is going to be

unpacked as Q of the states through time and the policy and the model parameters is again uh we see the likelihood of the model parameters and the likelihood of the policy coming out in front um and these are similar to the P but P doesn't have to equal q and then a slightly simpler uh piece here which is just about the states given the observations uh and so this is like literally given the retina Shadows what is the state estimate of the world so this is um it's like here in in P if we look at just this uh Second Line This is the probability of observations given States so this is given that it's daytime there should be a lot of photons if the estimated state of the world is nighttime there shouldn't be any photons $p(o|s)$ okay but it's O given s now in Q we actually have S given o so the p is different than the Q but also the orders have switched this is a called a recognition distribution because we're basically doing an estimation of the states given the observations so Q is kind of like the incoming way that the observations are going to be connected to the state estimators of the world that's the recognition but then the generative model which is the P that entails understanding what the hidden states of the world are s and your estimates of them that is and then generating out observations that are linked up to policy and so here there's a slightly different way that the parameters of

this model and that the um policy mapping is done with these uh few more statistical parameters but similarly which we don't need to go into too much detail in here is these are also going to be distributed using some statistical distribution that's going to help us estimate them and similarly we can take all the statistical parameters so for a normal distribution it's a two parameter model mean invariance but you could have uh distributions with one or with more than two variables but whatever number of statistical variables you want to estimate we're just going to wrap them into a function and figure out some way to estimate them and so that's a uh uh degree of Freedom you can think of in the model is that uh there's different you could flip out this normal for some other type of distribution or you could have another way to estimate the distributions uh normality okay that's p and Q but it is really important and interesting what they are any thoughts or questions or we can just keep cruising okay yeah um

Alex uh yeah I want to ask uh maybe in terms of uh so P is for generative model Q is for recognition distribution and recognition model and as we discussed in previous session generative model exists or acts uh in uh Dynamics so you have uh generative model in some way of it's like was example with Domino so when you act and act in accordance to this generative model and

an example of Road and a bike
this generative model defines way you
are looking on both ways and when go if
you don't see a buy and uh in this terms
so what will be recognition distribution
recognition model it's will be
understanding how bike is looks like or
what is how how road is looks like and
so on or it's the wrong direction for my
understanding a good question and it was
it was actually I had had to email Alec
to clarify a few things because I looked
at this Q distribution I thought wait
it's a recognition distribution but
where are the observations how can Q be
recognition when there's no observations
and then it turns out there are
observations that are uh related to q
but it's he just said basically it's the
norm in the recognition distribution
literature like this part of the basian
computational fields this is what they
call those two models the recognition
distribution which is from the
observations to the generative model
parameters which ironically as a norm
doesn't include the uh observation and
then the generative model again is the
one that's going from the organisms
parameter estimates of the world to
predicted observations and that's how
expectation maximization works now we
had all these interesting discussions
Alex which you're alluding to of for
example does the organism have the
internal model that's generative or is
the generative model enacted terms of
the relationship is it that's the

internalism versus externalism

discussion that we had so I'm going to try to combine what Alec inform me about as well as Channel a little Maxwell here and I think the answer is going to be there's multiple philosophical lenses that you could

apply um to real world situations and this is only what is specified in the math and in the programming and so if you include it if you include in your recognition distribution is the uh luminosity and the edge and the color and those are the observations and you're recogniz you're doing image classification on recognizing a bike I'm just simplifying it but there it is then that is strictly only what is in the recognition distribution it's going to be about the mapping of the observations um to of the states given observations and your observations could include those variables that we just talked about so if it's being conditioned on that then that's what's happening and then even given a single form ization of like okay the observations are the um overall brightness and the amount of movement or something like whatever it happens to be it could still be a a pluralistic philosophical scenario where somebody could say okay that's a modeling convenience but really those observations are in the space between the organism and the environment and someone else says yep it's a modeling convenience but I think those parameters are something about the neural firing

and someone else says it's both of them
and someone else said so I would be open
to
uh correction and also would love to
hear Alex perspective on this but I
think the answer would be these papers
that we read Tale of Two densities and
integrating externalism and internalism
are actually about how by Framing it
within active inference we move Beyond
those philosophical debates or another
way to say that is build a bigger tent
that includes both of those perspectives
um in a way that's compatibilist because
those are the debates that people had
about these models so like in the
reinforcement learning literature it's
like okay is the reward the signal you
know is it a pleasing you know aesthetic
body form is that the reward or is that
the signal of the reward and if it is
the reward then isn't that a circular
definition like organisms seek out
what's rewarding but it's not like you
just want to see the Aesthetics there
might be some second intention for
example and so how do you actually draw
out these questions is it happening in
the organism head with a reward or is it
enacted and that was sort of the
philosophical groundwork and now we're
for a minute putting those bags of all
different stripes down and looking at
just the way that we're going to specify
it and how it plays out in simulations
knowing that potentially one person
might have a basy and structural
computationalist perspective of these

examples we're going to look at in a second and somebody else might take a heavily inactive example they might say oh well the the cart that's on the hill it's like the cart Hill combo organism that's being or something like that but at the end whether the person thinks that it's inacted or it's just a basic and computationalist approach they're still going to at least be able to point to the same equations and ultimately the same code as well so that's like a hopefully explanation of how the active inference framework and the specifics end up unifying these General debates which potentially without such a unifying framework it'd be intractable cuz somebody would say okay well I think that Q is happening in the brain of the organism somebody else says I think that Q happens in between the niche and the organism someone else says I think Q is a modeling shorthand someone says I agree with person one and three or one and two and now we're saying okay they're kind of all correct they just all ways that we can phrase the specifics of what we're talking about here so not that everyone's correct because everyone's correct and there is no wrong answer but under this reconceptualization of action and inference we find that some of these previous questions are phrases in become like yes and instead of either/or okay to move a little bit more quickly just through these uh a few equations before we can I think just

highlight a little bit on the experiments and then probably return to the multiscale question that we talked about earlier and also about what other areas could control be interesting in and also what questions would we want to ask Alec and other attendees next week here is equation one and this I put after because I think it helps to understand that P and Q and the mappings but this is what is ultimately being minimized so again we don't get F that's the real free energy and that's um the one that we would really want to be converging towards which as they put it um converging towards an approximation of the posterior distribution of prob the P model of States policies and parameters given observations so that would be like the best possible State trajectory through time and the best possible policy given the observations of the world you know given that Bitcoin is at this value and ethereum is at this value the state trajectory that's the best for me entails this policy so that's clearly what we want to know but that is intractable so we needed to use some type of a convergence towards this intractability and this is where a lot of the more specific and technical details related to for example the collier informational Divergence the the informational distance between two distributions is related because um there's some those distance metrics are bounded at zero like you if two

distributions are identical their distance is zero and as they get more and more distant that number strictly increases and so it turns out that there's some ways to have like this hypothetical perfect distribution and then have a distance between where you're really at and this hypothetical perfect distribution which is always a positive distance and then try to converge towards that hypothetical perfect position by minimizing the distance function and that distance function is informational but it still is a distance metric in a certain way and wrote out a few details um the E is the expectation so it's expectation of States Through Time policies and model parameters and it's going to be something that combines the Q and the P the natural log which makes it a little bit easier to do the machine learning technique and a few other benefits scaling it and bringing certain values below the number line and then that is going to be always greater than or equal to that's this part right here than just the observed observations so it turns out that by doing this estimator we can bound um some a b there's a few things that get bounded but they're not even all phrased here

um yep so I the last thing I wrote there was given that the Q is um focusing on basically between p and Q you get all the parameters so what you can do by optimizing the recognition density and

the generative model back and forth
whether actually like a two cycle engine
like expectation maximization which is
how it gets implemented in code or
dynamically and continuously as in the
case of an active organismal Behavior
what has to get learned upon but here
what tractably can be learned on and
conditioned upon is that mapping of how
observations States and policies are
linked in a nuanced Nexus where they're
all taken into account rather than
running a classifier that goes from just
observations to States and then a second
layer model from states to policies that
would be a little bit more along the
lines of traditional machine learning
but this is why it's different
approach okay then um again we don't get
G uh we or we don't get F we we only get
G so here is another phrasing of f
that's being played out through
time so here is um building on three and
four we're looking at the variational
free energy not just for an instant here
this is like a kind of Timeless
understanding of free energy you don't
see T in here it's through time with the
S Tilda and the O Tilda but it's like
not a specific times and then this one
is actually a function through time this
is observation T so this is like that
specific observation and then here's
where we see the Divergence the D
distance Divergence the KL distance
between the Q and the p and those are
not usually exchangeable so like it's
not like driving between two cities so

this um section again you can unpack
some of what the equations actually say
but this learning in inference section
talks about how to update or uh do
inference do learning based upon the
data of these

parameters here is where we get to G so
we don't get F we only get to estimate G
so the way I kind of broke this down was
we we want to do more than just update
our internal model to bounce Sur prise
on observations so if all you wanted to
do was test retest accuracy on an image
classifier data set all you'd have to do
would be bound surprise on Cross
validated versions of that data um or on
uh uh subsample or testing and training
splits but we want to actually do a few
different things we want to reduce our
uncertainty in the future so that's to
the or the curs of T I don't know which
way You' say it um in terms of the
policies that we take now so it's quite
different than just minimizing our
surprise about a linear aggressor given
a static data set we want to reduce our
uncertainty about observations in the
future as a function of the policies
we're taking now and this minimiz this
tractable expected free energy function
 G takes as arguments policy and the
future time point so as a function of
the policies that are um available and
you know two years in the future how am
I going to um manage certain aspect of
life and it's going to be an expectation
that's conditional on um uh several
things conditional mainly we can

highlight is conditional on policy we're doing an expectation that's conditioned on policy because we wanted know something about the states of the world and the observations and the the estimated states of the world as well given a policy and then that is going to be strictly bigger um and so natural log that means that it's like basically worse um in some ways but it depends a lot on whether it's like positive or negative that's kind of confusing stuff sometimes um like whether it's a negative log or like the subtracting or the positive log and what it's always bigger than is the negative expectation of The observed states in the future given a policy me here's me observing that I have 50 Bitcoin um and that's this part two the P of observations given policy so what what I'd really like to know would be uh some type of model that just tells me what states I can expect in the future What observations I can expect in the future um which relate to the States because of the uh estimating the states given the observations but here we're going for the observations given the policy and it's the uh log and then uh this last part is kind of the important part which is the log of the generative model which is the observations given the policy that's how things will go that's well if we take this policy on miles per gallon then we will go between uh you know 1.5 and 2 degrees C now there's so many degrees of freedom that go into

that but that's like what we'd want to estimate would be relating how policies literally are going to play into observations which then tell us about states of the world estimated like S without a hat which should be hopefully related to S with a hat the real states of the world but still the focus here is on the observations that are conditioned on policy that kind of takes us to policy selection and they write that policy selections achieved by updating Q in order to re to minimize F via G and the uh need for this paper was that when you have a short temporal Horizon you can just run out every possible policy like every possible set of five chess moves or every possible set of all the Tic Tac Toe moves there can't be more than nine moves in a game every single step if the branching is small and the time Horizon is short you can evaluate G in full so it becomes rather than sampling over a distribution literally just calculating a table and then sorting however when you have deep temporal Horizons you can't really evaluate the full branching tree because there's too many options so you have to sample from that distribution so that's the sampling problem the second Pro and that's the Deep uh counterfactual sampling problem which is even difficult under uncertainty the second part is the Mario Kart Paradigm which is that in continuous action spaces there are infinite policies meaning that some other way is going to have to come into

play because not just are we sampling uh
deep through time we're sampling
potentially related to counterfactuals
which there's infinite counterfactuals
and we're talking about continuous
action spaces where there's act infinite
policies so if my only trading
affordances are should I have zero or
one or to bitcoin then I could ask which
one of those three options discreetly is
better but if there's infinite and I can
do continuous variables uh even smaller
than one you know breakdown it's like
then should I do

1.1 what if that's very different than

1.2 now you might say well those two
should be super similar and you should
be able just to sample every little
interval and estimate the distribution
right if it's a smooth distribution then
it's going to be easy to optimize maybe
you could discretize it maybe you could
do some other way to do smooth
optimization but the problem is actions
and policies are not smooth so maybe if
you're going into that turn in Mario
Kart you could uh aim at the corner and
just try to take it as fast as you can
or you could slow down and do a slightly
different angle or you could do some
other skidding maneuver so there's
multiple Separate Ways you're going to
be able to take this turn potentially
with different risks and reward and
they're all very nuanced so that is why
the Deep through time and counterfactual
aspects respectively uh which aren't as
much dealt with here counterfactuals but

through time absolutely is and then the continuous actions which is really the key introduction of this model that takes it out of the Paradigm of classifying images and puts it into the Paradigm of continuous control theory and then it turns out that that there's a lot of trajectories Through Time given how things can go and so there's some uh implementation of a way to sample trajectories that is utilizing what's called a Monte Carlo sampling approach or MC and uh there's the marov chain Monte Carlo sampling mc mc mc squared um and Monte Carlo just means that it's uh related to Monte Carlo being a card playing reference in English and it's like the way that you can shuffle a deck is so big how are you going to estimate the number of ways that you can get a royal flush or a pair of fours or something like that instead of just actually doing well it's 1 and $24 * 1$ and $52 * 3$ over 52 the way you do it is you just sample randomly and then you say okay I took a billion samples and 1 million of them had this feature and then you just take the ratio as your estimate so it's a sampling and that kind of hearkens back to earlier where we wanted to talk about getting the efficient sampling cuz that's kind of what it's about is in the marov um or sorry in the Monte Carlo sampler which could be marov chain based um we want to get good sampling we don't want to sample only the hands with a royal flush like if we had a a poor randomizer

because then we're going to get an estimate that's wrong all right then they go into a little bit more detail about the G the estimated or the expected free energy which is still as it was before about policy in the future and now it's going to be uh this broken down and this is also where they draw some of the homologies to reinforcement learning and info gain into the uh the informational gain kind of like the reward and the info so this is where the explore exploit also comes into play which is that if one is only optimizing locally they may be exploiting they may be getting reward for a short period of time but then it doesn't lead to a um uh like we'll see it in the hill climber example but if you just reward only and only exploit sometimes you don't do very well but if you just explore you might not ever get to exploit anywhere good and so the whole tradeoff and the whole challenge of modeling is this balance between explore and exploit between searching deeply and searching broadly and so the Holy Grail would be the right way to search deeply down the routes that are interesting and informative but not down the ones that don't tell you that much but we don't know which paths are going to be informative or not until we go down them however if we have a deep generative model we can say in terms of me winning this chess game this branch is not informative for me to search down not because there aren't trillions of options but because I have

two castles and they have no pieces other than their king so I'm going to win so in terms of me winning it doesn't give me any information to keep on searching down that route I already know I'm going to win and then there might be another aen where there's fewer options per se like less branching but there's more uncertainty that's resolvable more info gain is possible with respect to the policies and winning all right let's just do a little bit on the experiments before we close out so the two experiments that they go into so hopefully all of you experimentalists um by Sasha um hopefully all you experimentalists will like this part and the two examples are the mountain car and the uh inverted pendulum task and the hopper task so the mountain car is an example of exploration so what's being learned is the policy between position and velocity and then getting higher up the reward function is like getting higher up and so the car doesn't have a strong enough engine just to drive up to the flag so it's got to drive up and then the perfect policy would be at right when you're stopped flip the engine into reverse and keep it in reverse until you stop on this side and then get a little bit higher up this way but this shows you why you can't locally optimize why you need to explore because let's just say you're driving uphill you're slowing down you're slowing down you start slipping back slipping back and you're

like no stay on the gas continue
accelerating forward now you're slowing
your descent so when you get to here
you're not moving that fast so then when
you're on this side you'd say no no I
still want to get to the flag and so
you'd end up just sort of like being
very shallow down and focusing a lot of
your time here on the um middle and on
the right side so if you just simply
move towards the flag you're probably
not going to get there if the engine is
too weak however if you actually say I
want to explore I want a policy that
helps me increase my range so I want to
go as far as I can here and then wow if
I enact a policy where as soon as I hit
my Maxima I just try to explore further
this way and then I enact a policy
Switch and Go back to going this way wow
I might actually reach the goal purely
as a function of exploring now again
this is like a little model situation
where it turns out that exploring does
help one realize the goal that's not
always the case but it's it isolates
that feature of the model needing to be
able to explore the second case in the
exploit paradigms it's usually a little
bit more obvious what needs to be
exploited and about whether you're doing
it well or not so in the inverted
pendulum task it's like a railroad car
on a rail and there's a heavy ball M
that's on top of this pendulum it's
inverted and then the car can like move
backwards and that kind of moves this
joint to the left and that pushes the um

Pendulum In this case would straighten it out whereas if this car starts taking off to the right then this pendulum is just going to fall and so the reward is the height of the pendulum so it's trying to maximize its reward by enacting a policy and so the policy that's best would be to keep on exploiting that perfectly straight up line where you actually are in the zone of optimal control and then the exploit task on the right is about uh increasing the distance that's traveled by a little jumping thing does anyone else have any like thoughts on explore exploit or um what what else does that yeah goad blue so this I thought was super interesting um as I was unpacking this paper so and maybe I should just kind of like lay a little bit of this out there so the reward is only given in this extrinsic value which is like the exploit portion of the free energy equation like given in number six and so you know I don't know you didn't say it explicitly but like here in the exploration experiments the reward just uses that um exploit parameter and doesn't do any exploration based on state or parameter Information Gain and then in the exploit task it only uses that exploitation parameter so like there's no need for exploration I just thought that that was super interesting and like what would happen if they included exploration in this explor in the exploitation um Paradigm so like why did

they leave that out like it was just
kind of weird to me cool good question
like that makes me think about let's say
there's some parts of the railroad track
where it's different angles and so yes
you want to maintain the pendulum being
upright but also you might want to
explore around a little bit to find a
region where you can control it better
so that actually makes me think of like
doing a track stand on my bicycle
there's some times where there's some
slopes of pavement that are easy to do a
track stand on and some are really hard
to do it but the way you get from a hard
position into an easier part of payment
you have to explore and that means
breaking the track
so definitely real control tasks are
related to the combination of explore
and exploit why did they use these two I
think that the reason why and let's hear
from Alec as to the real answer is that
these are part of the open AI
benchmarking set of like little micro
computable benchmarks for learning
algorithms here's the open ey AI um
let's see if we can even open it here
yeah so like here's their website and uh
there's a bunch of uh documentable cases
that allow comparison between different
machine learning algorithms and so I
think they're just outlining the minimal
single dimensional like sort of extreme
classic test cases there's a ton of
cases where we can already see it
wouldn't work like a double pendulum
that's chaotic or there's other kinds of

things that this model just doesn't even try to do but this allows us to directly compare two other algorithms on the exact same task instead of inventing something that would like either not take advantage of active inferences strength or be like kind of an unfair example by making it too nuanced like if we added in a reward another type of something that a generative model could take advantage of then we couldn't compare it to some of these other agents that don't use that kind of information is that I just think they were using the minimal case to to demonstrate what explore and exploit could look like and then future work hopefully would be about things that involve both right and so I get that and but it just was like interesting to me then I guess because there's no reward in the exploit parameter that that's why they had to keep the exploit parameter in the explore task do you know what I mean like what I I just wonder what would be the results if you took out the exploit parameter from the explore task would it then not succeed because it just would wander around aimlessly or would wandering around aimlessly or like without a reward to get higher would that just produce like a random result do you see what I mean I I see what you mean so there there's there's one there's one driving strategy where it's truly just making policy that's unrelated to its elevation and um that depending on how that plays

out it might get somewhere it might not most likely not because the space of the strategies with relationship to getting the flag is low but yes I I I I agree that's an interesting question like when is it enough to just explore like when is novelty search simply enough when might novelty search be strictly the best when might novelty search 80% plus 20% let's keep a little reminder of our elevation when might that be a relevant and tractable strategy which parts of these explore and exploit are calculable um for example it might make sense to evaluate or exploit a current distributional uh value but how do we explore the space of economic strategies like I can't explore every single possible economic strategy but I might be able to exploit some ratio by doing Arbitrage so there's some control tasks where the exploitation is really straightforward and visible and the exploration is hard and then there's other ones where um maybe this car can't see the flag so maybe exploring is the only thing it can do and so the really great question those are kind of interesting about these things being traded off so there's the explore exploit tradeoff and how that gets implemented there's also the model adequacy minus complexity which is we want a good model but not too complex or overfit one so there's no a prior best way to trade off those two and then um there there's just all these interesting trade-offs that we can explore the space

of let's just look at what the results were so here the x-axis is like where positions is from the other side farther to um 0.5 which I guess could be the flag uh or at least close to it I try to line it up maybe it's the way or something like that but we can see that the position of the greedy agent who just drives up up up up up and says no and then just goes up up up up no this one stays tightly bounded in uh a medium velocity uh regime and within a sort of sampling a ton from this part of the distribution think about that like an upside down bell curve and it's like sampling a ton from right here but it's not sampling from up here so that's how there's a relationship between sampling and control because to get to certain States it's like sampling those States the Epsilon uh greedy uh agent which has a slightly different implementation it does sometimes appear to get outside maybe by the end of its 100 EPO it was starting to take some trips out further but and at slightly higher uh speeds and slightly further out but it doesn't spend most of its time doing so the active inference learner of course they wrote the paper so it's going to be the one that succeeds but it does in 100 epox learn very fast fast and it does get to some pretty high speeds and some pretty distant positions and then just briefly to like compare it with the other task the two exploit tasks are laid out here and they are comparing it to this

ddpg agent which is a model-free learner and so people can look into details or one could imagine that maybe there's other algorithms they could test it against these are all the kinds of things we can ask AI about but we can see that um a which is the same as c c is just the first part of a it just zoomed in on really close but it's like right away in both of these after either like five epochs like five trials five seconds or um after maybe 30 or 40 or 50 the active inference learner just starts to take off in terms of once it's hit on a solution sometimes it just it strictly improves it with some noise whereas the um

ddpg the two things that you see are first off it spends a ton of time near the Baseline so it takes a while before it locks on to any useful combination of parameters and then when it gets off Baseline it ends up um sometimes succeeding but still um it's like perhaps because it's locally optimizing or we can ask why it is but they're just not sampling highly rewarding States even after it gets off the Baseline it does stay off Baseline but it doesn't ever get as high as the active inference learner does extreme ex rapidly so this is kind of where they're drawing their evidence for the claims about the potential superiority or preference for active inference now yes you could probably train a ddpq on some massive data set and get better performance than they got they probably just used a

straightforward implementation so that's where research happens that's what people study is okay well can we do a ddpg whatever that is plus this other feature that is being used by this other group or what is it that actually causing the active inference learner to get off the ground so fast and could we even get it off the ground faster in the case of this Hopper task or could we make a more reliable reward function for the inverted pendulum task but this is control theory at its finest really and so this is what it's all about is learning policies that map states onto reward those are the key pieces that's why we spent so much time talking about the equations that are relating between observation

State estimates and policies because that's what it's all about is the observations coming in and then the policies being selected based upon the estimated states of the world and then actions coming out that's the markov blanket of the organisms perspective at least and then outside uh World in these examples is simple and unchanging like gravity is gravity for the pendulum and for the hopper and it's not changing but we can start to think about well what if there was uh a dial gravity Rhythm how would the control theory change well there could be a generative model in the agent control uh mapping of those p and Q that involved some sort of is it day or night and then I have a conditional policy whether I think it's day or night

so we can kind of see how it starts
adding layers there and yeah that's
basically it but let's have any last
thoughts does anyone like what was
something cool or unexpected what's
something they'd like to ask Alec about
or that they're curious to learn more
about

now chill Oh Shannon go
ahead hey um yeah so I was thinking
about um Blue's question and why not use
the um exploration component in these
inverted pendulum and Hopper tasks and
wouldn't it be because there's kind of
only one state that is maximum reward so
if we're completely vertical or if we
hop as far as possible but if there were
potential states that would have
differing values of reward then we might
expect to see something different if we
use the the exploration component of the
active reference model great Point like
if it were a modal distribution then you
can't just try to keep it as close to
the maximum height as possible because
you wouldn't know if you were locally
optimizing or globally optimizing and
you might want to be sampling across the
whole distribution just so you know or
you might want to spend a lot of time at
rewarding States but still have some
understanding of the Total distribution
to understand if there might be more
rewarding states that you're just not
visiting at all that is the whole
challenge of local and Global
optimization and so you're right that in
this mountain car example local

optimization is global you're going to get to the best flag just by by you know so to speak I know it was was in the explore strategy but in this one it's clear as you brought up if you're having the pendulum at the maximum height You're simply doing as well as you can period you don't need to explore other ways you could do Loop to Loop and that's the challenge though is you got to you got to walk through the valley of death to get from the local Optima to the global and that's where these algorithms uh dare not tread because they have to walk through improbable or non preferable like moving away from your target States in order to get to the next Peak which might be absolutely higher in the exploit Paradigm or might reveal new territory in the explore Paradigm so there's reasons why you want to get through the valley of low likelihood whether you're exploring or exploiting or a combination and that is where there's so many options there's so many weeds and tangles in the valley that the way that we're going to cut through it with a machete is infog gain and we're going to explore paths through the valley that are the most informative for us given our understanding of our own policies and so that's like if you have a 4x4 versus a small team versus you're a drone different policies different affordances mean that you're going to be exploring this Valley differently and then there's whether you're exploring

for the sake of exploring and learning
new turf or whether you're exploiting
because you just want to have uh high
elevation or whether it's a little of
both those are all some Nuance you can
add in but the inside of active
inference is like let's actually think
about what would be informative
trajectories to take through that Valley
based upon a generative model of the
landscape and taking our own policy
affordances into account if you don't
have a generative model of the landscape
you're going to be doing something like
locally just going downhill or locally
going uphill so you need a deeper
generative model of landscape and if you
don't take your affordances into account
then you're lost because you might be
unknowingly optimizing something that's
going to be appropriate for you know
know the uh drone to fly over but then
you're on the ground so you actually
can't make it up that hill but it's a
policy that might work for a different
organism and it's just not the one that
works for you I know we're kind of like
mixing metaphors but I think it's a fun
way to think about these optimizations
because we are implementing Control
process something that we modeled with
control theory and then also we're
seeing it play out in like a very
abstract computational framework
here cool stuff metors yep so then they
talk about the relationship with
previous work I think I'll just put it
on the screen people can um pause it if

they want to look but with that I'll
just say thanks to our cohort here for
participating super interesting stuff
and next week we hope to have Alec on to
discuss some of these questions so you
can check the calendar event to provide
a follow-up survey where you can add
some details how could we improve or
what questions would you like to hear
conveyed in 8.2 and and for those who
are not participating live please
provide us with feedback or suggestions
or questions that we can cover and just
uh stay in touch but that was really
great so thanks a lot everyone for the
discussion